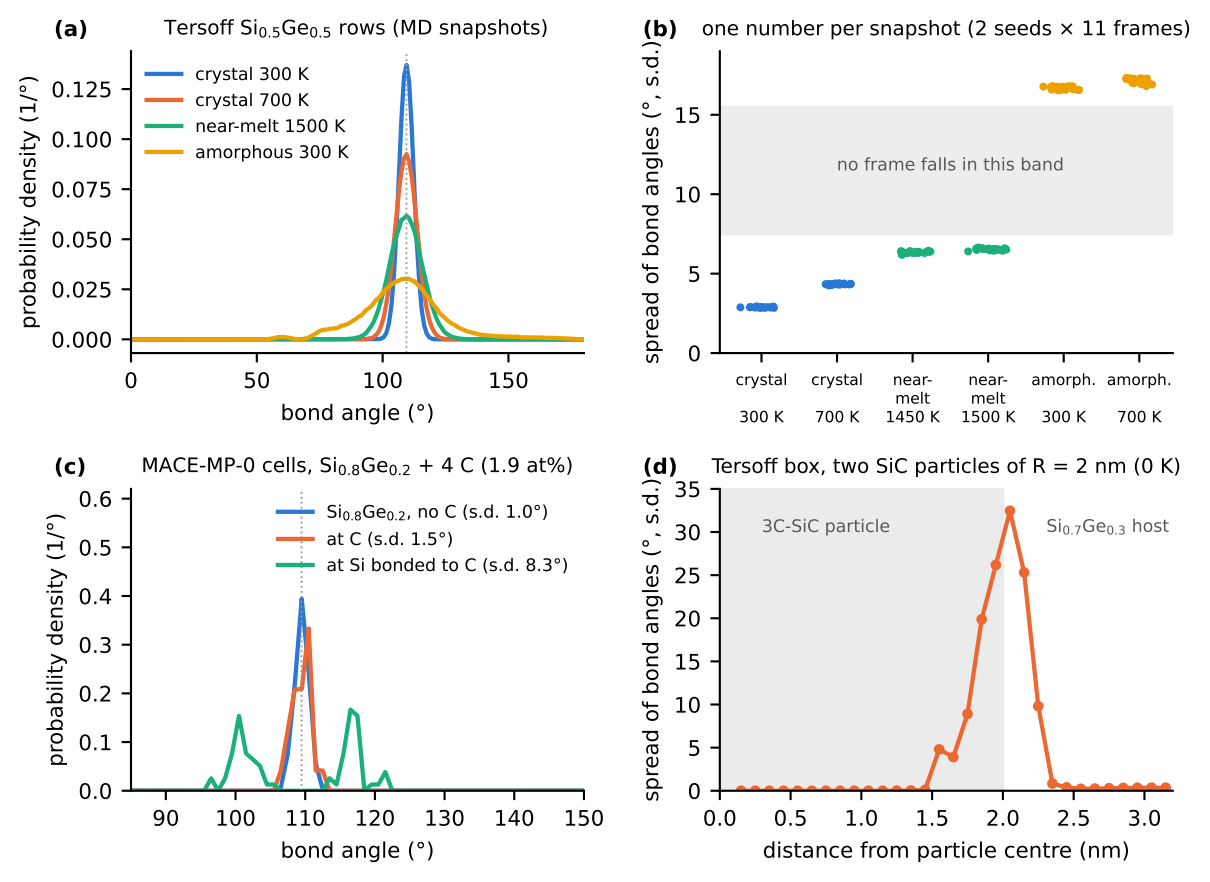
(a)
Tersoff Si0.5Ge0.5 rows (MD snapshots)
crystal 300 K
crystal 700 K
near-melt 1500 K
amorphous 300 K
probability density (1/°)
0.125
0.100
0.075
0.050
0.025
0.000
0
50
100
150
bond angle (°)
(b)
one number per snapshot (2 seeds × 11 frames)
spread of bond angles (°, s.d.)
no frame falls in this band
15
10
5
0
crystal 300 K
crystal 700 K
near-melt 1450 K
near-melt 1500 K
amorph. 300 K
amorph. 700 K
(c)
MACE-MP-0 cells, Si0.8Ge0.2 + 4 C (1.9 at%)
Si0.8Ge0.2, no C (s.d. 1.0°)
at C (s.d. 1.5°)
at Si bonded to C (s.d. 8.3°)
probability density (1/°)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
90
100
110
120
130
140
150
bond angle (°)
(d)
Tersoff box, two SiC particles of R = 2 nm (0 K)
3C-SiC particle
Si0.7Ge0.3 host
spread of bond angles (°, s.d.)
35
30
25
20
15
10
5
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
distance from particle centre (nm)